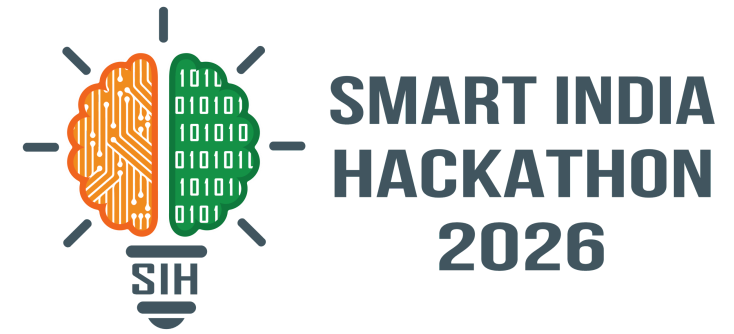




SMART INDIA HACKATHON 2026



- **Problem Statement ID** – 26087
- **Problem Statement Title** - *AI-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem*
- **Theme** - *Smart Education*
- **PS Category**- *Hardware*
- **Team ID** – 26087
- **Team Name** – *HexaForge*



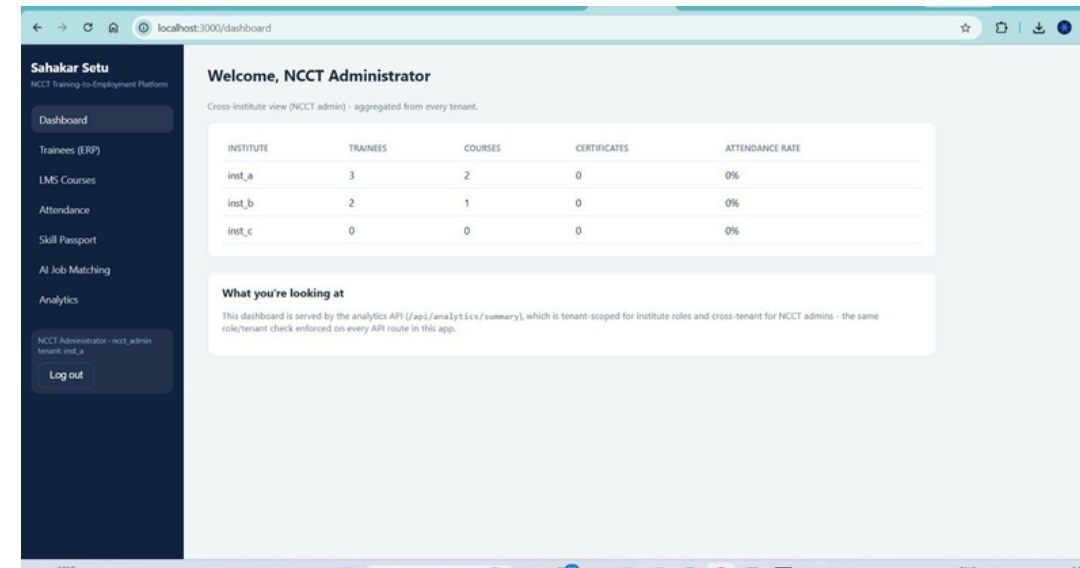
Proposed Solution

NCCT lacks a **unified, AI-enabled platform** for training, attendance, certification, and employment across its institutes.

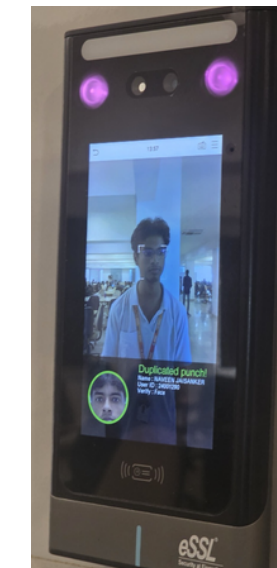


- Develop an **integrated platform** combining **Training ERP, LMS, attendance, certification and employment services**.
- **Store and manage** trainee, programme, assessment and certification data in a **centralized database**.
- Provide **multilingual and mobile-friendly learning** with assessments and progress tracking.
- Create a **Digital Skill Passport** with verified skills and certificates for every trainee.
- Use AI-based career guidance, job matching and analytics to **connect training outcomes with employment opportunities** and institutional monitoring.

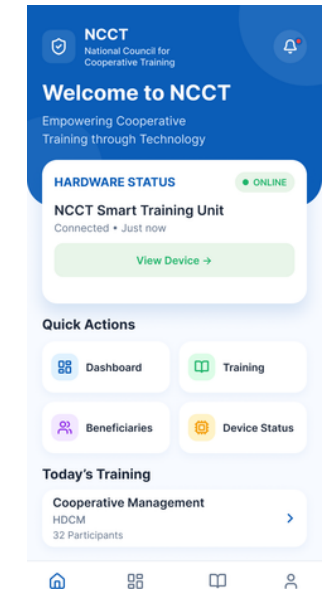
Prototype



Web Dashboard



Attendance



Mobile application

Key Capabilities

Training ERP

Manages registration, nominations, batches, trainers, attendance, timetable, hostel and logistics.

Smart LMS

Provides multilingual courses, interactive content, assessments and progress tracking.

Digital Skill Passport

Stores verified skills, training records and QR-based digital certificates.

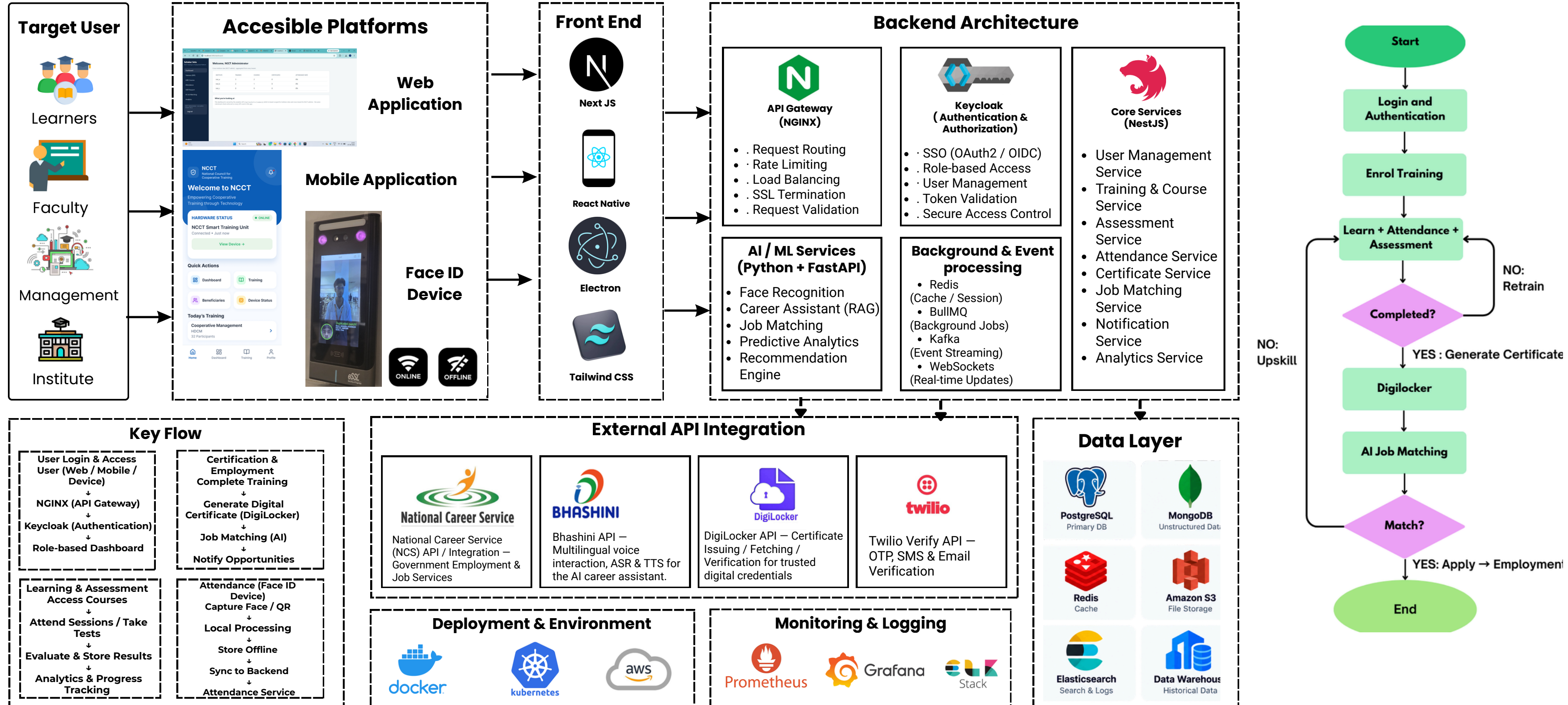
AI Career & Job Matching

Provides personalized career guidance, identifies skill gaps and connects trainees with suitable employers.

Analytics Dashboard

Tracks training performance, certifications, employment outcomes and institutional progress.

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

Feasibility

1

Technical Feasibility

Uses proven technologies like AI, cloud, mobile apps, QR/face recognition, and Raspberry Pi hardware.

2

Operational Feasibility

Supports NCCT's multi-institute structure with centralized records and offline-first access for rural users.

3

Economically Feasibility

Uses open-source technologies and low-cost hardware, reducing deployment and maintenance costs.

4

Regulatory Feasibility

Compliant by Design — Aligns with DPDP Act 2023, data privacy, accessibility, and data-localisation requirements.

VIABILITY

SCALABILITY

Modular design enables expansion across NCCT institutions, programmes, users and regions.

MARKET VIABILITY

Connects trainees, institutes, and employers for training-to-employment.

SUSTAINABLE VIABILITY

Reduces costs and supports long-term scalability across NCCT institutes.

FINANCIAL VIABILITY

Reduces administrative costs while improving training efficiency, employability and institutional resource utilization.

01

DATA INTEGRATION

Connecting ERP, LMS, certification, and employment systems.

02

SYSTEM SCALABILITY

Supporting multiple NCCT institutes and large user volumes.

03

BIOMETRIC PRIVACY

Secure handling of face data and consent.

04

USER ADOPTION

Designing for low digital literacy and multilingual users.

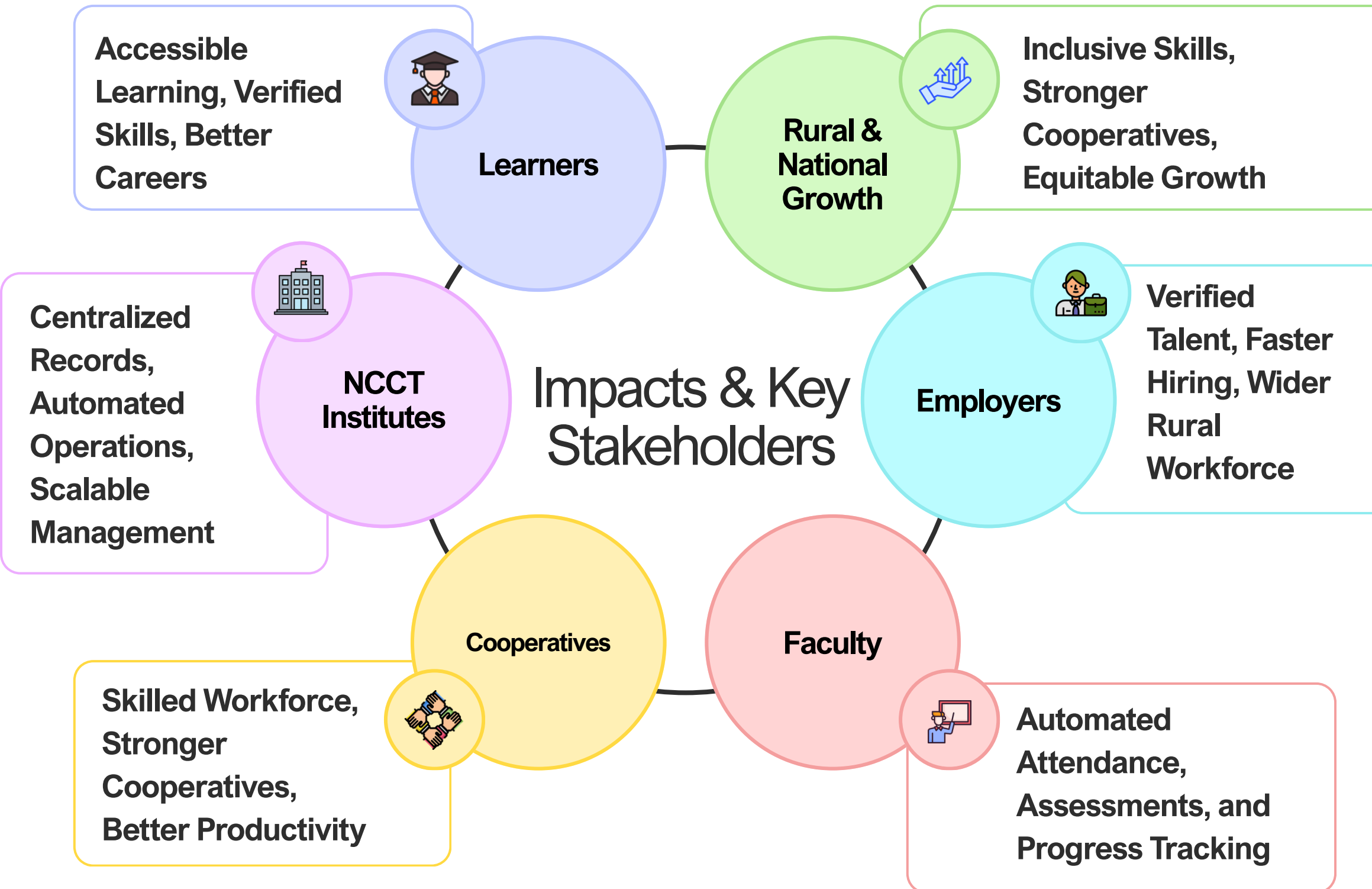
05

OFFLINE DATA SYNCHRONIZATION

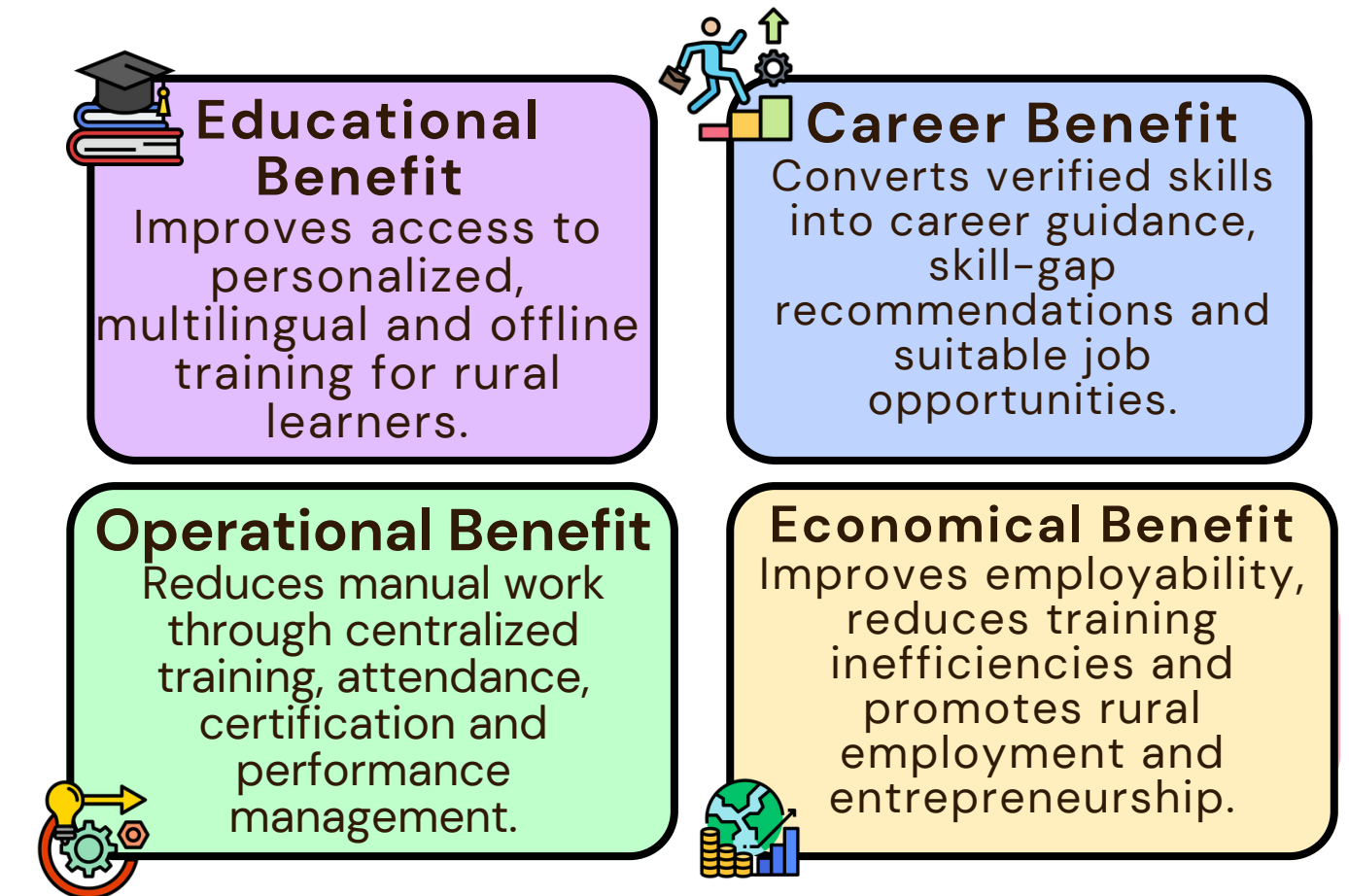
Maintaining consistent data during intermittent connectivity can create synchronization conflicts.

IMPACT AND BENEFITS

Impacts



Benefits



Driving Sustainable Impact

Our solution directly supports key UN Sustainable Development Goals by enabling inclusive education, employment, innovation, and rural development.



1. Gogoi, A., Manoranjini, M., & Gupta, M. (2025). Design and implementation of digital literacy training programme: Findings of a quasi-experimental study from rural India.
 - Source: PLOS Digital Health
 - Link: [Research Paper](#)
 - Summary: Evaluates a digital literacy training programme in rural Jharkhand and demonstrates how structured digital training, accessible technology, and locally relevant learning can improve digital skills and access to educational and career opportunities. This supports SahakarSetu's rural-first, multilingual and low-resource training approach.
2. Xiao, Y., & Li, Q. (2026). Bayesian network modelling for predicting employment tendency and major matching of vocational education students.
 - Source: Discover Artificial Intelligence, Springer Nature
 - Link: <https://link.springer.com/article/10.1007/s44163-026-00952-7>
 - Summary: Proposes an AI-based framework that uses academic performance, skills, career interests and labour-market demand to improve career recommendations and reduce skill–job mismatches. This supports SahakarSetu's AI-based career guidance and job-matching module.
3. Kaur, D. H. (2025). Offline Realities in an Online Era: Barriers to Virtual Learning in Rural Indian Colleges.
 - Source: International Education and Research Journal
 - Link:
 - Summary: Examines barriers to online learning in rural India, including unreliable internet, limited devices, electricity issues and digital-literacy gaps. The findings support SahakarSetu's lightweight, offline-first approach, where learning content can remain accessible without continuous internet connectivity.

